Market Microstructure Effects of Algorithmic Trading on Price Discovery

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Abstract

This research explores whether high-frequency algorithmic trading (HFT) strategies improve market quality under average conditions while amplifying volatility, becoming harmful to retail investors during periods of market stress. Through a systematic literature review of peer-reviewed studies published between 2010 and 2026, this paper analyzes the relationship between HFT activity, price discovery, liquidity provision and retail investor outcomes. Findings indicate that while HFT compresses bid-ask spreads and accelerates price adjustment under stable market conditions, it systematically withdraws liquidity during stress events (evidenced by the 2010 Flash Crash) creating a structural speed and information asymmetry that disadvantages retail participants. This paper concludes that the benefits of HFT are unevenly distributed and argues for targeted regulatory interventions, including minimum order resting times and enhanced circuit breaker mechanisms, to protect retail investors without eliminating the efficiency gains HFT provides.

Keywords: high-frequency trading, price discovery, market microstructure, retail investors, liquidity, flash crash, algorithmic trading

1. Introduction

High-frequency trading (HFT) now accounts for approximately 50% of equity trading volume in U.S. markets and 30% in European markets, granting algorithmic systems an outsized influence over modern market quality (Lin et al., 2024). While HFT is recognized as a tool employed by sophisticated financial firms to capture marginal profits at scale, its broader implications for market stability and retail investor outcomes remain deeply contested. HFT is an automated system designed to execute trades within milliseconds, profiting from fractional spreads that accumulate into substantial capital gains. Research has documented a meaningful correlation between HFT activity and improvements in liquidity, particularly through compression of the bid-ask spread, though the direction and magnitude of these effects are highly sensitive to macroeconomic conditions (Yacoubian, 2025).

Retail investors (individuals trading personal capital using publicly available analysis tools) occupy a structurally disadvantaged position relative to algorithmic participants. Operating with smaller capital bases, higher brokerage fees, and decision-making subject to behavioral biases, retail investors are particularly exposed to the adverse conditions HFT produces during market stress. The 2010 Flash Crash, in which the Dow Jones Industrial Average plummeted nearly 1,000 points within minutes, stands as the most documented case of HFT-amplified market disruption and serves as the central anchor for this analysis (Aliyev et al., 2024).

This paper presents a systematic literature review examining the mechanisms through which HFT affects price discovery and retail investor outcomes. Data collection was restricted to peer-reviewed studies and institutional reports published between 2010 and 2026, requiring

methodological transparency, minimal stated bias, and direct relevance to the research question. The central hypothesis guiding this research is that HFT imposes measurable negative consequences on retail investors — particularly during stress regimes — consequences that existing regulatory frameworks have insufficiently addressed.

2. Literature Review

A. Foundational Theory

Three foundational frameworks establish the theoretical basis for understanding how algorithmic trading affects price discovery: the Efficient Market Hypothesis (EMH), the Kyle (1985) Model, and Market Maker Theory. The Efficient Market Hypothesis holds that asset prices fully reflect all available information, making it extremely difficult to consistently outperform the market. EMH operates in three forms: weak (prices reflect past trading data), semi-strong (prices adjust to all public information), and strong (prices reflect all information including private knowledge) (Corporate Finance Institute, n.d.). The emergence of HFT raises a central question within this framework: whether algorithmic speed accelerates convergence toward efficiency or introduces distortions that undermine it.

The Kyle (1985) Model describes a market with three participant types: an informed insider trader who knows the true value of an asset and trades strategically; noise traders whose randomized trades inadvertently obscure insider activity; and market makers who set prices based on observed order flow. Together these participants create a dynamic in which information is gradually incorporated into prices: a process directly relevant to understanding how HFT, with its speed and informational advantages, interacts with and potentially distorts this equilibrium (Kyle, 1985). Market Maker Theory extends this foundation by establishing that liquidity is not a free public good but a service rendered at cost, with the bid-ask spread representing its price. The emergence of HFT displaced traditional human market makers with algorithmic equivalents, raising the critical question of whether algorithmic market making is conditionally withdrawn under the precise stress conditions when liquidity is most needed.

B. Empirical Evidence

The empirical literature on HFT presents a consistent pattern: algorithmic trading improves measurable market quality under normal conditions while becoming a source of fragility during stress events. Yacoubian (2025) found that HFT-dominated equities maintained a bid-ask spread of 0.8 basis points compared to 1.5 basis points for non-HFT securities (a compression of nearly 47%) representing a meaningful reduction in transaction costs under stable conditions. However, intraday volatility rose from 0.45% in normal periods to 1.1% during stress events (an increase of 144%), with liquidity provision by HFT firms declining measurably during elevated-volatility windows. Lin et al. (2024) corroborate this, documenting that HFT amplified volatility by 30% to 40% in already-volatile markets, as synchronized algorithmic responses to the same signals create selling pressure that human participants cannot counteract at equivalent speed.

The BIS study by Aliyev et al. (2024) provides long-run structural evidence: while average bid-ask spreads have declined across equities, bonds, and foreign exchange markets over three decades, the skewness of liquidity distributions has increased in equity and bond markets. This means typical liquidity conditions have improved while extreme liquidity deterioration during stress has become more pronounced; structural breaks in liquidity skewness consistently coinciding with macroeconomic shocks. Their framing captures the dynamic precisely: liquidity has become more efficient on average and more fragile at the extremes.

C. Gap and Contribution

Despite a growing body of literature documenting both the liquidity benefits and volatility risks of HFT, a meaningful gap persists. The academic literature has largely examined HFT's effects on institutional market quality metrics without centering the retail investor as the primary unit of analysis. The result is a knowledge asymmetry that mirrors the market asymmetry itself: HFT firms possess a comprehensive understanding of algorithmic trading's structural effects, while retail investors (the ones most exposed to its adverse consequences) operate without equivalent awareness or regulatory protection. Furthermore, existing studies examine HFT's benefits and risks in isolation, without a unified framework accounting for the regime-dependent nature of its effects. This paper addresses both gaps by applying a conditional liquidity framework that unifies the contradictory findings in the literature and extends to concrete regulatory proposals.

3. Results and Discussion

A. The Flash Crash and Conditional Liquidity

The Flash Crash of May 6, 2010 represents the defining empirical case for the conditional liquidity hypothesis. The Dow Jones Industrial Average, NASDAQ, and S&P 500 collectively shed approximately one trillion dollars in market value within minutes before partially recovering. The joint CFTC-SEC investigation determined that a large institutional sell order of approximately 75,000 E-mini S&P 500 futures contracts created the initial imbalance. Rather than absorbing this order flow as traditional market makers were obligated to do, HFT firms rapidly bought and immediately resold contracts, generating volume without net liquidity provision, then withdrew entirely as conditions deteriorated (Kirilenko et al., 2017). The resulting liquidity vacuum allowed prices to fall far beyond levels justified by any fundamental reassessment of value, confirming that HFT-provided liquidity is not a reliable public good but a conditional service that disappears under stress.

The strongest counterargument holds that HFT merely amplified a crash caused by institutional behavior, not algorithmic failure and that withdrawal during stress reflects rational risk management. This argument has merit but conflates cause and amplification. Even if HFT did not initiate the Flash Crash, the CFTC-SEC report (2010) and Kirilenko et al. (2017) establish that HFT behavior measurably deepened and prolonged it. In a market where HFT constitutes 50% of volume, the distinction between cause and amplification is operationally secondary to the scale of harm produced.

B. Retail Investor Harm

Retail investors bear disproportionate HFT-related harm through two distinct channels. The first is speed asymmetry: HFT firms exploit microsecond execution advantages to anticipate and trade ahead of slower retail order flow, extracting value from participants who cannot match their technological capabilities (Menkveld, 2016, as cited in Yacoubian, 2025). The second is infrastructure cost pass-through. Lin et al. (2024) document that processing $10,000 in equity transactions requires data from approximately 535 billion quotes and 35 billion trades. Volume that imposes substantial infrastructure costs on exchanges and data brokers, ultimately passed downstream through higher brokerage and data fees that fall disproportionately on retail participants. Retail investors are simultaneously disadvantaged in execution quality and charged more for market access; the BIS study further documents that increased liquidity skewness means they are most likely to encounter extreme illiquidity precisely when they most need to transact (Aliyev et al., 2024).

C. Policy Implications

The conditional liquidity framework points toward targeted regulatory intervention rather than prohibition. Three measures are warranted by the evidence. First, minimum order resting times (requiring limit orders to remain active for a minimum duration before cancellation) would directly curtail flickering quote strategies identified by Lin et al. (2024), reducing information asymmetry without eliminating legitimate market-making. Second, stress-calibrated circuit breakers should trigger not only on price movement thresholds but on liquidity withdrawal indicators (specifically the rate at which HFT firms cancel orders and reduce quote depth) intervening before feedback loops become self-sustaining, as supported by the BIS liquidity

skewness findings (Aliyev et al., 2024). Third, mandatory disclosure of order-to-trade ratios would provide regulators with data to identify manipulative behavior (a measure already implemented in modified form under MiFID II in Europe) without broad prohibitions that sacrifice HFT's demonstrated liquidity benefits.

4. Conclusion

This paper examined whether high-frequency trading improves market quality under normal conditions while becoming a source of volatility and disproportionate harm to retail investors during stress. The evidence supports this thesis comprehensively. HFT demonstrably compresses bid-ask spreads, accelerates price discovery, and reduces average transaction costs, but these benefits are conditional. They are produced during stable market conditions and evaporate precisely when stress creates the greatest need for stable liquidity provision.

The 2010 Flash Crash remains the defining empirical case, but the BIS study's documentation of increasing liquidity skewness across global markets suggests it is not an anomaly but a symptom of structural fragility that has grown as HFT has come to dominate market microstructure. Retail investors bear this fragility disproportionately through speed asymmetry and infrastructure cost pass-through, a double burden existing regulatory frameworks have not adequately addressed. Minimum order resting times, liquidity-calibrated circuit breakers, and mandatory order-to-trade ratio disclosure represent concrete, targeted interventions that preserve HFT's efficiency contributions while constraining its capacity for harm. These measures do not require the elimination of algorithmic trading, rather its accountability. As HFT systems increasingly incorporate machine learning and artificial intelligence, the regulatory gap identified here will only widen if not addressed proactively. The market of the future will be shaped by whoever designs its rules today, and those rules must account for what happens not only when markets are calm, but when they are not.

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